Errors Are Not Choke Points: poori research ek kahani mein

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Simple Hinglish guide. Ise padhne ke baad paper (`paper/main.pdf`) ek baar padho, sab samajh aa jayega.

1 Shuruaat: problem kya tha?

LLM (jaise Qwen, Llama) maths ka sawaal step-by-step solve karta hai. Isko **chain of thought (CoT)** ya **trace** kehte hain. Kabhi beech mein ek step galat ho jaata hai aur final answer galat aa jaata hai. Poori field ka common idea yeh hai:

1. Pehla galat step dhundho (isko **first error** kehte hain; human annotators isko mark karte hain).
2. Wahan tak wapas jao aur wahan se dobara generate karo (**backtracking / repair**).

Process Reward Models (PRMs) isi kaam ke liye train hote hain, aur ProcessBench jaise benchmarks isi ko grade karte hain: "kya tumne pehla galat step pakda?" Hamara original aim tha **early failure detection**: kya hum trace ke shuru mein hi bata sakte hain ki yeh fail hoga, taaki jaldi intervene kar sakein?

Phase	Kahani ek nazar mein
1. Early detection	detector bana, par woh sirf sawaal ki difficulty padh raha tha (negative result)
2. Naya sawaal, v_k	failure kahan decide hota hai, model ke apne continuations se mapa
3. Main result	error $\neq$ choke point; repair, restart ko same compute par nahi harata; theory se proof
4. ChokeBench	naya benchmark; check kiya ki yeh asli recovery predict karta hai
5. RAG (secondary)	hallucination par same idea; result grader par depend karta hai, human rating baaki
6. Review aur refocus	do reviews ke baad claims narrow kiye, kamzor hisse imaandari se likhe
7. Run D	reasoning models (Qwen3, R1) par exact on-policy test: theory sahi, failures zyaadatar budget khatam hone se

2 Phase 1: early detection, aur pehla jhatka

ProcessBench (3,400 solutions, 12 models, har solution mein human-marked first error) par humne detectors banaye. Number achhe lag rahe the, lekin phir humne ek simple **baseline** lagaya: sirf sawaal ka text aur kaunsa model hai, trace ka ek bhi step dekhe bina.

Detector ko kya diya	AUROC	Matlab
Sirf sawaal + model (koi step nahi)	0.588	sawaal kitna mushkil hai, bas itna pata
+ trace ke pehle 25% steps	+0.005 (n.s.)	early steps se lagbhag kuch nahi milta
+ poora trace	+0.031	thoda signal, par tab tak bahut der ho chuki

Seekh (negative result)

Early detector asal mein *"sawaal mushkil hai"* padh raha tha, *"yeh attempt fail hoga"* nahi. Isko **difficulty confound** kehte hain. Yahi hamara pehla pivot tha: naya sawaal poocho.

3 Phase 2: naya sawaal, aur ek hi measurement v_k

Naya sawaal: **failure asal mein kahan "decide" hota hai, aur kya wahan se repair karna restart se behtar hai?** Iske liye ek quantity:

$$v_k = P(\text{sahi answer} \mid \text{sawaal} + \text{pehle } k \text{ steps})$$

Isko estimate karne ka tareeka: trace ko step k par kaato, *usi model* se wahan se 8–16 baar aage likhwao (**continuations / rollouts**), aur gino kitni baar answer sahi aaya. Total 125,832 continuations chalaye (Kaggle ke free T4 GPUs par).

Term	Simple matlab
v_0	bilkul shuru se solve karne ka chance. Yahan se dobara likhna = **restart**.
Backtrack to step j	step j se just pehle tak rakho, wahan se dobara likho (yaani v_{j-1} wala point).
Choke point (decisive step)	woh step jahan v sabse zyada girta hai. Yahan failure asal mein decide hota hai.
First error	woh step jahan human kehta hai "yahan galti hui".
Capability-limited	$v_0 \leq 1/8$: model ke liye sawaal shuru se hi lagbhag namumkin.

Example: $v = [0.6, 0.6, 0.1, 0.0]$. Step 2 par v 0.6 se 0.1 gira, toh choke point step 2 hai. Agar human ne step 3 ko galat mark kiya, toh *error* aur *choke point* alag hain.

4 Basics: choke point kaise dhundhte hain (step by step)

#	Kya karte hain	Detail
1	Failed trace lo	ProcessBench solution, steps mein toota hua (step 1, 2, …, n), final answer galat
2	Har prefix banao	$k = 0$: sirf sawaal. $k = 1$: sawaal + step 1. … $k = n - 1$: sawaal + pehle $n - 1$ steps
3	Usi model se aage likhwao	har prefix se m baar (8–16) continuation, temperature ke saath (har baar alag)
4	Grade karo	har continuation ka final answer gold answer se match? Haan = 1, nahi = 0
5	v_k nikalo	v_k = sahi continuations / m. Trace fail hua tha, isliye $v_n = 0$
6	Drop nikalo	step j ka drop = $v_{j-1} - v_j$ (step j likhne se chance kitna gira)
7	Choke point	jis step ka drop sabse bada, wahi choke point
8	Pehle filter	agar $v_0 \leq 1/8$: model shuru se hi fail, koi choke point nahi ("capability-limited")

Worked example ($m = 8$, 4 steps, human ne step 4 ko galat mark kiya):

Prefix k	0 (sirf sawaal)	1	2	3	4 (poora trace)
8 mein se sahi	5	5	4	1	–
v_k	0.625	0.625	0.500	0.125	0
Step j ka drop $v_{j-1} - v_j$		step 1: 0	step 2: 0.125	**step 3: 0.375**	step 4: 0.125

Choke point = **step 3**, human error = step 4. Yahi "errors are not choke points". Step 3 ne chance 0.5 se 0.125 gira diya, jabki human ko galti step 4 mein dikhi. Step 3 se pehle ($v_2 = 0.5$) se resample karna, restart ($v_0 = 0.625$) se behtar nahi hai. Yahi Proposition 1 average mein kehti hai.

Noise ka kya? 8 samples se v thoda noisy hota hai, isliye teen safety checks: (i) *Bayesian posterior*: har step ke liye "yeh choke point hone ka probability", sirf ek winner nahi; (ii) *sample splitting*: aadhe samples se choke point chuno, baaki aadhe se score karo; (iii) do alag batches ke v curves match karte hain ($r = 0.985$). **PRM choke point kaise guess karta hai?** PRM har step ko 0–1 score deta hai. "Largest drop" rule: jahan PRM score sabse zyada gire, woh step. Isko hum ChokeBench par check karte hain. **Run D mein farq:** wahan steps nahi, *token positions* par cut (0, 256, …, 5120), kyunki reasoning models ke traces lambe aur bina clear steps ke hote hain.

Zaroori shabd	Asaan matlab
On-policy	continuation *usi* model se jisne trace likha; tabhi v_k us model ka asli chance hai
Restart	$k = 0$ se dobara; har attempt ka success chance v_0
Matched compute	sabko barabar GPU budget; PRM chalane ki cost bhi gini. "4×" = 4 restarts jitna budget
P(solved) in N tries	$1 - (1 - p)^N$; ek point jiska p zyada ya attempts chhote ho, woh budget mein jeetega
AUROC	0.5 = coin toss, 1.0 = perfect. Detector fail/pass ko kitna alag kar pata hai
95% CI [a, b]	asli value lagbhag is range mein; agar range mein 0 hai toh fark pakka nahi (n.s.)
p-value	itna bada fark sirf luck se aane ka chance; < 0.05 = significant
Bootstrap	data ko baar-baar resample karke CI nikalna
Holm correction	kai tests ek saath, toh false alarm rokne ke liye p ko adjust karna
Preregistration	plan aur hypotheses data se pehle git mein commit; baad mein badal nahi sakte
Cohen's κ	do graders kitna agree karte hain (0 = random, 1 = perfect; 0.2 = kamzor)
Spearman ρ	do rankings kitni match karti hain (-1 se 1)
Supermartingale	"fair game": fail hone wale khel ka beech ka state, average mein, shuruaat se behtar nahi hota

5 Phase 3: kya mila (main results, maths)

Sawaal	Jawab (number)	Matlab
Kitne failures step 1 se pehle hi decide?	33–66% (4 models)	repair karne ko koi step hi nahi
Baaki mein human error = choke point?	0.438–0.647 (chance 0.16–0.20)	chance se behtar, par aadha hi sahi
Naye model (Llama-3.1-8B) par confirm?	haan, 0.633	preregistered test, pass
PRM kya pakadta hai? (ChokeBench)	human error 73%, choke point sirf 45%	verifier label pakadta hai, asli point nahi
Choke point ke liye train kiya model?	39%	better nahi hua
Restart kitna bachata hai? (4 attempts)	44–66% failures solve	simple restart bahut strong
PRM backtracking vs restart (same compute, Llama)	-0.063 (confirmed)	repair, restart se *haara*
Free perfect oracle vs restart (4× budget)	sirf +0.028	jaadu ki chhadi bhi bas thoda jeetti

Matched compute ka matlab: har method ko barabar GPU budget (tokens, PRM ki cost sab gini gayi). Tabhi comparison fair hai.

Kyun? (theory, Proposition 1: supermartingale)

Ek failed trace ka koi bhi point, jo aage dekhe bina chuna gaya ho, average mein shuru se behtar nahi hota: $E[v_\tau \mid \text{fail}] \leq v_0$. Simple bhasha: *jo trace fail hua, uska aadha hissa ek fresh start se zyada keemti nahi hota*. Isliye choke point dhundhne se tokens bachte hain, success nahi badhti. (Proposition 2: iska fayda bas chhote attempts hai, budget ka lagbhag 1.25–1.4×.)

6 ChokeBench: hamara naya benchmark

ProcessBench poochta hai "pehla galat step pakda?". **ChokeBench v1** (230 failed traces, 4 models) *dono* poochta hai: human error pakda, *aur* choke point pakda? Aur woh step resampling ke liye kitna keemti hai? Sabse important check: **kya ChokeBench score asli recovery predict karta hai?** (circular na ho, isliye samples ke aadhe hisse se choke point label kiya, baaki aadhe se gain mapa.)

Check (exploratory)	Result
Named step = choke point: recovery gain mein badhat	+0.40
Named step = human error, choke point ko control karne ke baad	+0.01 (n.s.)
Naye (held-out) model par ranking, Spearman ρ: ChokeBench vs ProcessBench-style	+0.52 vs +0.22

Matlab: human label tabhi kaam aata hai jab woh choke point se match kare. ChokeBench asli fayde ko behtar predict karta hai.

7 Phase 4 (secondary): hallucination wala RAG study

RAG mein model ko ek source diya jaata hai aur answer source se hi aana chahiye. Source mein jo nahi hai woh likh diya toh **hallucination**. Data: RAGTruth (human labels), model Llama-2-7B. Detector D = LettuceDetect (har sentence ko score karta hai).

Method	Cost (base = 1×)	Judged hallucination change vs base (E1 judge)
DRA: detect, restart, abstain	1.36×	-0.042 (exploratory)
CARE (choke-point-aware controller)	1.77×	-0.047 ($p = 0.0003$, preregistered)
CARE vs plain restart	–	-0.005 (koi fark nahi)
Chain-of-Verification	3.75×	mehenga, aur behtar nahi

Proposition 3 (law): restart loop jo answer deta hai, uski hallucination rate = detector ka $P(\text{hallucinated} \mid \text{pass})$, retries kitne bhi ho. Retries sirf coverage badhaate hain. Human labels se prediction: 50.2% → 26.6%. Check: 4× retries par coverage +2.4 points, hallucination lagbhag same (+0.6 se +1.5). Prediction sahi nikli.

Imaandari: yeh part kamzor kyun hai

(1) Judge E1 humans se kam agree karta hai ($\kappa = 0.219$; 81% drafts flag karta hai jab humans 50%). (2) Grader badlo toh result badal jaata hai: detector jaisa grader +0.326, independent judge +0.017 (n.s.). Gate khud ko grade kar raha tha. (3) Outputs ko abhi tak humans ne rate nahi kiya. Isliye RAG *secondary* hai, aur hum "hallucination solve" ka claim **nahi** karte.

8 Imaandari ke kadam (reviewers yahi dekhte hain)

Kya kiya	Kyun
Preregistration	hypotheses data se pehle git mein commit kiye, taaki baad mein result chun na sakein
PRE / EXP / CONF tags	har claim ke saath likha: pehle se plan tha, baad mein dekha, ya naye data par confirm hua
Sample splitting	ek half se chuno, doosre se score karo; warna restart nakli tareeke se achha dikhta
Bootstrap CIs, Holm correction	har number ke saath uncertainty; kai tests ki galti control
Mistakes withdraw kiye	"dual judge" aur "1,500-item" table galat the, hata diye, corrections log mein likha
Blinded human kits	base vs DRA ke 80 pairs, 2 raters; key ka hash pehle commit, key repo se bahar
External review	do reviews ke baad focus narrow kiya: main claim maths wala, RAG secondary

9 Run D: naya bada experiment, aur uske results

Kyun? Pichhle traces ProcessBench ke the, kisi aur ne likhe the, isliye sirf "near on-policy" the. Proposition 1 exactly on-policy traces par hold karti hai. Aur aaj ki field ka sawaal *reasoning models* ka hai (jo hazaaron tokens sochte hain, `<think>` ke andar khud backtrack karte hain). Run D yahi test karta hai.

Setting	Value
Models	Qwen3-1.7B (thinking mode), DeepSeek-R1-Distill-Qwen-1.5B
Data	MATH-500, problems 21–500 (480; pehle 20 sirf pilot ke liye, analysis se bahar)
Stage 1	har problem ka ek trace, max 6,144 tokens; math-verify se grade
Stage 2	har failed trace ko 9 jagah kaato: 0, 256, 512, 1024, 1536, 2048, 3072, 4096, 5120 tokens (paragraph break par snap, taaki cut sirf past par depend kare); har cut se $m = 8$ continuations
Hardware	Kaggle, 2× T4 GPU, vLLM; har model ka ∼6.5 ghante ka session

Hypothesis (preregistered)	Test
D1: Proposition 1 exact on-policy par	har cut t par $v_t - v_0 \leq 0$? Agar kisi cut par significantly positive (Holm $p < 0.05$), toh theory refute
D2: koi cut restart ko 5 points se nahi harata	same token budget par best cut (half A se chuna, half B par score) minus restart; 4× par upper CI $< +0.05$
D3: descriptive	kitne failures shuru se decided ($v_0 \leq 1/8$), aur v sabse zyada kahan girta hai
D4: sensitivity	D1, D2 bina un traces ke jo token limit par kat gaye

Kya hua: Qwen3 ka stage 1 pura hua, lekin uska stage 2 T4 par bahut slow (∼92 tokens/s) tha aur time limit mein ek bhi batch pura nahi hua. R1 ke 40 failed traces ka stage 2 pura hua (10.5 million tokens).

Result	Matlab
Accuracy: Qwen3 70.8%, R1 72.7%	dono models 480 mein se lagbhag 70% sahi
Failures mein budget khatam: Qwen3 95%, R1 92%	reasoning models zyaadatar *galat step* se nahi, *sochte-sochte budget khatam* hone se fail hote hain. Wahan dhundhne ko koi galat step hi nahi
D1 (R1): $v(t) - v(0)$: $t = 256$ par 0, $t = 5120$ par -0.212	**theory sahi nikli**: failed trace ka koi cut fresh start se behtar nahi
D3: 60% failures capability-limited	purane result (33–66%) jaisa
D2 preregistered: best cut − restart = +0.133 [+0.067, +0.197]	rule ke hisaab se restart *haara*. Lekin yeh fragile hai (neeche)
D2 plug-in check: +0.031 [-0.045, +0.114]	significant nahi

D2 fragile kyun: per attempt koi cut restart ko nahi harata (D1). Fayda sirf *saste attempts* se aata hai: late cut par budget kam bachta hai, toh ek attempt restart ka lagbhag 1/5 cost karta hai, aur 4 samples se ∼19 attempts ka andaza lagaya jaata hai. Jeffreys prior 0/4 success wale cut ko bhi ∼10% chance deta hai. Imaandar reading: reasoning models par late resampling *shayad* saste attempts se pay kare (Proposition 2 ise allow karti hai), lekin hamare samples kam hain. Paper (Sec. 8) mein dono numbers likhe hain.

10 Ant mein: ek line mein research

Final takeaway

Errors are not choke points. Jahan human galti dikhata hai, failure aksar wahan decide nahi hota; bahut failures shuru se hi decided hote hain; verifiers label pakadte hain, asli point nahi; aur us point se repair karna, same compute par, simple restart ko nahi harata. Isliye restart woh baseline hai jise har localise-and-repair method ko harana chahiye. Hamara naya ChokeBench yeh test karne ka tool hai.

Abhi baaki: (1) zyada samples ke saath Run D ka D2; (2) human ratings (base vs DRA, 2 log, 2–3 ghante each); (3) doosre researchers ka replication. **Paper padhne ka order:** Abstract → Intro ke 5 contributions → Sec. 5–7 (counterfactual, policies, ChokeBench) → Sec. 8 (Run D) → Sec. 9–10 (RAG) → Limitations → Appendix (proofs).